

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Poster Presentation

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO3 –Poster Presentation	Assessment:	Assessment Tool 1– Poster Presentation
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	S.Sirivally	Reg. No.:	192472371
Section / Batch:	2024- CSE(AI)	Date:	18-08-26

Code of Conduct

I _S.Sirivally / 192472371_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __Sirivally.S__

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	



RAINFALL PATTERN ANALYSIS USING CONTINUOUS PROBABILITY DISTRIBUTIONS



Exploratory analysis of monthly and annual rainfall across Indian districts

Guided by:
Dr. Krishnaraj M

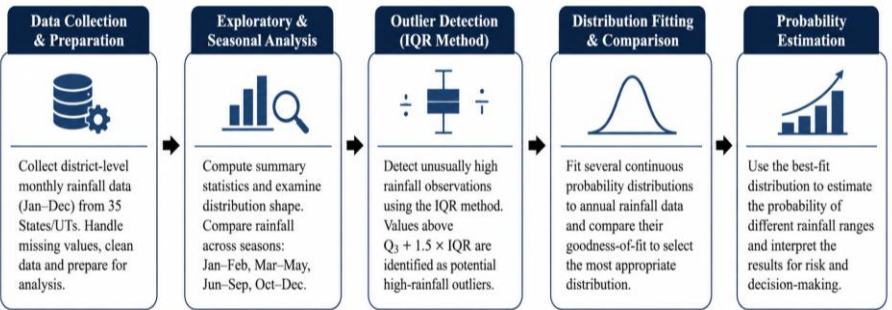
Submitted by:
S.Sirivally(192472371)
Department-CSE(AI)

AIM & OBJECTIVES

To analyze rainfall patterns across Indian districts and estimate rainfall probabilities using continuous distributions.

- Analyze rainfall patterns across Indian districts and seasons.
- Summarize rainfall using statistical measures and distribution analysis.
- Identify unusual rainfall values using the IQR method.
- Use continuous probability distributions to estimate rainfall probabilities

METHODOLOGY



KEY FINDINGS

- Annual rainfall is strongly right-skewed.
- June–September is the dominant rainfall period.
- IQR highlights unusually high annual-rainfall observations.
- High variability exists across districts.

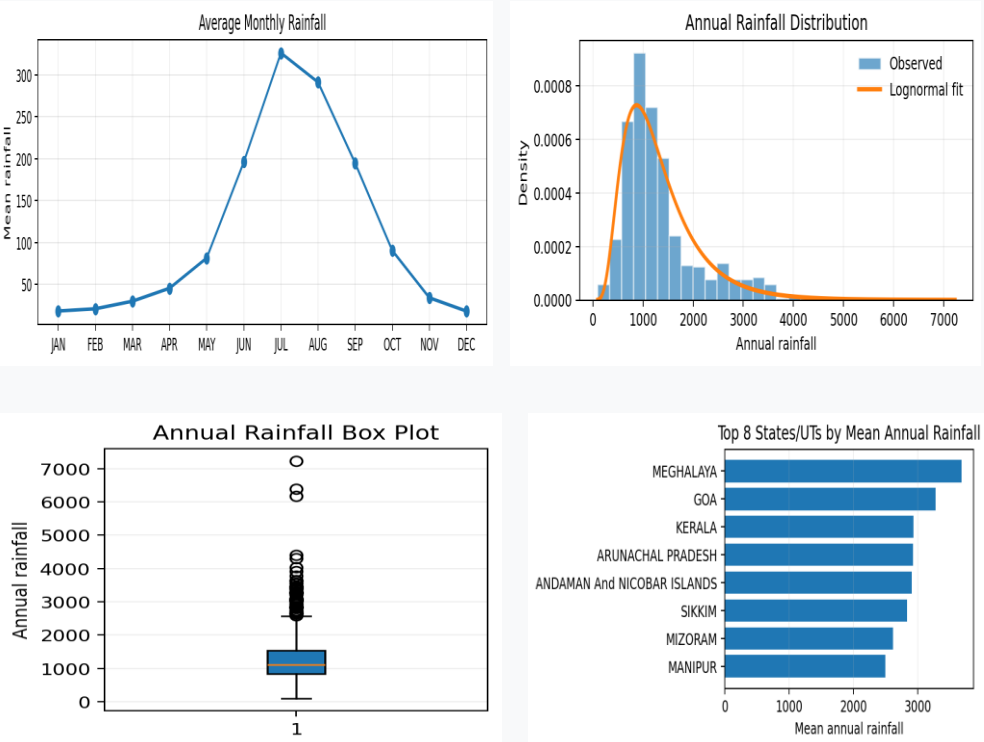
DATASET

Indian District Rainfall Dataset

641 district records | 35 States/UTs | Rainfall in mm
Variables include State/UT, District, Jan–Dec rainfall, seasonal totals and annual rainfall.

Source: Kaggle dataset

RESULTS



STATISTICAL INTERPRETATION

Continuous probability view

- Rainfall distribution is not symmetric.
- Skewness = 2.19, indicating strong positive/right skewness
- A small number of high-rainfall districts create the long upper tail.
- The fitted distribution helps convert rainfall observations into probability-based estimates.
- These estimates can support rainfall-risk and resource-planning decisions.

CONCLUSION

Indian district rainfall is highly variable, seasonal and strongly right-skewed. The combination of descriptive statistics, IQR analysis and probability distributions provides a useful framework for understanding rainfall variability and estimating rainfall likelihoods.

FUTURE SCOPE / ENHANCEMENTS

- Add multi-year records to study temporal change.
- Compare districts using regional probability models.
- Evaluate more distributions and goodness-of-fit tests.
- Develop drought and extreme-rainfall alerts.
- Integrate climate and geographic variables.

1347.0

Mean annual rainfall

1116.2

Median annual rainfall

2.19

Skewness